Three Small Puzzles

an intuition kit for *Order Without a Centre*

working notes

October 2026

Each puzzle is small enough to do with a pencil. Each has the same shape: three things, three pairwise reports, every report fine on its own, and the question whether one picture of all three can stand behind them. Do the checks by hand before reading the commentary. The last question in each section is for you, not for the mathematics.

1. Three watches (gauge)

Setup. Three friends compare watches two at a time, in whole minutes.

- A and B: "B reads exactly what A reads."
- B and C: "C reads exactly what B reads."
- C and A: "A reads one minute *more* than C."

Check 1. Show that no three settings a, b, c satisfy all three reports. (Go round the loop and add the offsets: $0 + 0 + 1 \neq 0$.)

Check 2. Now nobody knows the absolute time: each watch's reading is "anything, evenly spread". Each pair's report is then perfectly consistent with each watch on its own being evenly spread. Read only the *parity* (even or odd minute) of each watch. Verify by listing the eight parity patterns that

$$[\text{A and C same parity, B the other}] \;=\; \tfrac{1}{2}[A{\equiv}C] - \tfrac{1}{2}[A{\equiv}B] - \tfrac{1}{2}[B{\equiv}C] + \tfrac{1}{2}.$$

The reports say $[A{\equiv}B]$ and $[B{\equiv}C]$ always hold and $[A{\equiv}C]$ never does, so the question on the left would have probability $-\frac{1}{2}$. The contradiction is *visible*.

Commentary. This is a *gauge* twist: renamings (offsets) that do not compose back to zero. A coarse reading (parity) catches it. If you may only ask how watches stand to each other, one pair at a time, it cannot be caught. Round the rotating Earth this is real: the loop misses by about 207 ns, and GPS fixes it by choosing a centre.

For you. Where do you meet offsets that each pair agrees on but that do not add up round a loop (schedules, exchange rates, conventions)? Who, if anyone, plays the centre?

2. A tournament (order)

Setup. Players P, Q, R. P beats Q, Q beats R, R beats P.

Check 1. Show no ranking fits. (Each report is a direction; the three directions close into a cycle.)

Check 2. Try to build a question about *all three* from questions of the forms "does X beat Y?" and "is X ranked at the top?", using only "not" and "either one" (joining questions that can never both be yes). Notice that "P beats Q" and "Q beats R" can both be yes, and so can almost any two such questions about different pairs. "Either one" never gets to join them, so every question you build stays about one pair. With three or more possible

scores this is a theorem [proved; core Lean-checked]. So comparisons, even with each player's own score, can never pose a question that exposes the cycle.

Commentary. This is an *order* twist: directions that close into a loop. With finitely many scores that wrap round, a joint record of a pair (the two scores together) exposes it, as in the watches. With scores running off to infinity and answers only yes or no, nothing askable exposes it at all [Lean].

For you. Where do you meet rankings that are each defensible pairwise but form a cycle? Were the records you saw pairwise outcomes, individual scores, or joint records of pairs?

3. Three committees (relational)

Setup. Committee X chooses one of 3 options, committees Y and Z one of 4. Each pair of committees reports only which combinations of their choices are *acceptable*, nine each, and that every acceptable combination happens equally often ($\frac{1}{9}$ each).

X–Y acceptable	X0: Y0,Y1,Y2; X1: Y0,Y1,Y3; X2: Y1,Y2,Y3
Y–Z acceptable	Y0: Z1,Z2; Y1: Z0,Z1,Z3; Y2: Z0,Z2; Y3: Z2,Z3
Z–X acceptable	Z0: X0,X1; Z1: X0,X2; Z2: X0,X1,X2; Z3: X1,X2

Check 1. The pair reports agree where they overlap: X is spread evenly; Y has shares $(2,3,2,2)/9$ from both reports that mention it; Z has $(2,2,3,2)/9$.

Check 2. Nothing is *logically* impossible. There are 16 triples whose three pairs are all acceptable:

$$(0,0,1)(0,0,2)(0,1,0)(0,1,1)(0,2,0)(0,2,2)(1,0,2)(1,1,0)(1,1,3)(1,3,2)(1,3,3)(2,1,1)(2,1,3)(2,2,2)(2,3,2)$$

written (X,Y,Z), and every acceptable pair occurs in at least one of them.

Check 3. Yet no plan (no weighting of the 16 triples) produces the even $\frac{1}{9}$ spreads. Give each acceptable pair a score:

X–Y	+1: X1Y1, X1Y3, X2Y1, X2Y3; −1: X0Y1, X1Y0, X2Y2; others 0
Y–Z	+1: Y0Z2, Y2Z2; −1: Y0Z1, Y1Z3, Y2Z0, Y3Z2, Y3Z3; others 0
Z–X	+1: Z0X0, Z1X0; −1: Z0X1, Z1X2, Z2X0; others 0

For each of the 16 triples, add the scores of its three pairs: every total is ≥ 0. (Example: $(0,0,1)$ has X0Y0 = 0, Y0Z1 = −1, Z1X0 = +1, total 0.) So any plan gives the scores a non-negative average. But the reports give every acceptable pair weight $\frac{1}{9}$, and the scores sum to −3, so the average would be $-\frac{1}{3}$. No plan fits.

Commentary. This is a *relational* twist: many-to-many acceptances whose frequencies do not fit together, though their possibilities do. It is the smallest case where every question you can ask, built from the pair reports by "not" and "either one", stays consistent, so the contradiction is *hidden* [no plan: hand certificate above; every question stays consistent: float computation, exact check owed]. With fewer options on two of the links it cannot hide; this size is exactly where it first can, and the threshold is a known one from the study of contingency tables (Sullivant 2006), and so is the equivalence; only the reading as "checks you can run" is ours.

For you. Where do you meet "these combinations are acceptable" between pairs of parties, with no preference among them? Did anyone ever check whether the pairwise arrangements could be one arrangement?

What to carry from the three

- *Gauge:* bookkeeping that does not close; caught by a coarse joint reading, missed by relations alone.
- *Order:* directions that close into a loop; hidden only at infinity.
- *Relational:* acceptances whose frequencies do not fit; hidden as soon as two of the relations are fine enough.